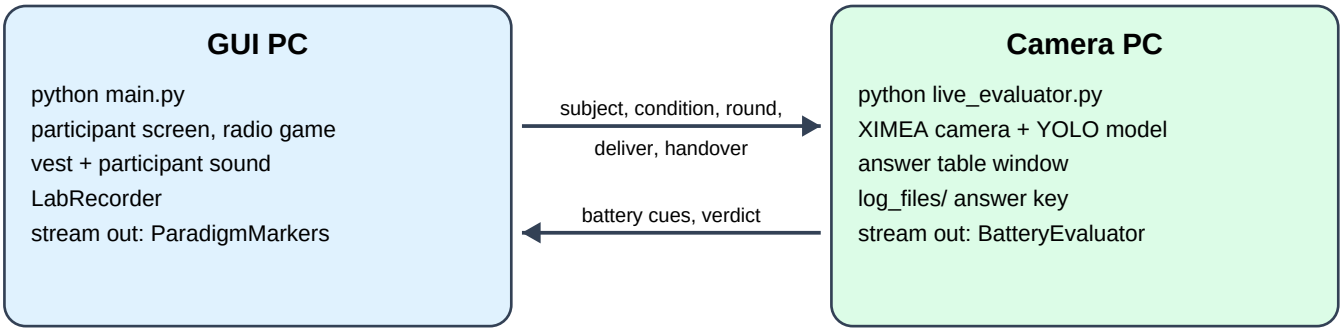
Model + GUI connection: usage guide

How to run the dual-task GUI together with the camera evaluator (YOLO model) on two PCs, what happens in a round, and what to check when something does not work. Everything here matches the code in this repository; it has been tested with simulated data, so treat the timing values as starting points to tune on the real setup.

1. How the two PCs talk



same local network (cable recommended) - LSL finds the other PC by itself

The link is LabStreamingLayer (LSL). LabRecorder only *records* the two streams; it is not needed for the connection itself.

2. One-time setup

Both PCs

- Clone / pull this repository (same version on both PCs - the camera PC needs `log_files/`, `config.py` and `sequences.py`).
- Same local network, by cable if possible. Wi-Fi works but adds jitter.
- Windows Firewall: allow Python on **private** networks on both PCs, otherwise LSL cannot see the other PC.

```
pip install -r requirements.txt
```

GUI PC

- bHaptics Player running and the vest paired (vest cues go through the Player).
- Participant headphones connected (device name containing `Trekz`; change in `main.py` if you use another one).
- `USE_REMOTE_EVALUATOR = True` in `config.py` (default). While it is True the manual y / c keys are off.

Camera PC

- Model libraries (not in requirements.txt): `ultralytics`, `torch` for that PC's GPU, and the XIMEA Python API.
- Model path and camera settings are at the top of `live_test.py` (`BASE_DIR`, `MODEL_PATH`, `CONFIDENCE`, exposure).
- The model must have the classes `tray`, `hand`, `Station` and the four battery colours (Blue, Red, Yellow, Green).

3. Check the link once (2 minutes)

Do this after any network or firewall change, before the participant arrives.

Step	Where	Do	You should see
1	GUI PC	`python main.py`	Setup screen.
2	Camera PC	`python live_evaluator.py --no-camera`	Console: `[LSL] Connected to GUI markers on ...`

Step	Where	Do	You should see
3	GUI PC	Pick a subject and a condition, start it.	Camera PC console follows: subject, condition, answer file, then RADIO GAME / ASSEMBLY with the countdown.
4	Camera PC	Ctrl+C	Link is fine. If step 2 or 3 fails, see Troubleshooting.

4. Running a session

Order	Where	Do
1	GUI PC	Start bHaptics Player, connect vest and headphones, then `python main.py`.
2	Camera PC	`python live_evaluator.py`. Check the bottom-left says **GUI linked** and the camera image shows the green/grey **EVALUATION AREA** on the station (not STATION NOT FOUND).
3	GUI PC	Start LabRecorder, select `ParadigmMarkers` and `BatteryEvaluator` (plus your other streams), start recording.
4	GUI PC	Setup screen: subject, condition, game mode, language - Start. The camera PC loads `log_files/sXX_<condition>.txt` for that subject by itself.
5	Both	Run the condition (8 rounds). Then scoreboard, NASA-TLX, back to the main menu. Pick the next condition; the evaluator keeps running.
6	GUI PC	After all conditions: choose `sus` in the condition menu for the experience questionnaire and the subject summary files.

Start order does not matter much: either program waits for the other and reconnects if one is restarted. If you start the evaluator *after* the setup screen was confirmed, tell it the subject and condition once:

```
python live_evaluator.py --subject 3 --condition audio
```

5. What happens in one round

Moment	Camera PC	GUI PC / participant
Checkpoint, 5 s retrieval	Shows RETRIEVAL countdown.	Participant picks up the piece.
Assembly starts	Row of this round turns green: evaluation active. Expected answer comes from the log file and is checked against what the GUI shows.	Battery task appears, timer runs (20 s; fast 15 s, slow 25 s).
Each battery placed	Battery seen in a slot (first frame if the model is sure, else 2 frames): check or cross in the table, message `Eval_Battery_...` sent. A hand at the station does not block this.	Cue of the condition (table below). Marker `Battery_Cue_...`
Round ends when the tray leaves the station (off it for 10 frames, no hand at it). Not at the third battery, not when the clock runs out, not at "deliver".	Whole tray judged: correct only if all 3 slots match; a slot not seen counts as wrong. Marks turn black, Result shows OK / MISTAKE. `Eval_Verdict_...` sent.	Verdict applied to score and log (`Assembly_Evaluated_...`, source camera).

Cue per condition (per battery)

Condition	Correct battery	Wrong battery
audio	positive_sound2.wav	current mistake sound (loud_incorrect_answer.wav)
haptic	vest positive 3 (warm swell, back)	vest negative 2 (sharp staccato, front)
audio_haptic	sound + vest	sound + vest
baseline, training, robot_fast, robot_slow	nothing (marker only)	nothing (marker only)

Sounds are cut to 1 s when played. One cue per battery: a battery that was only covered by a hand does not cue again; one that is taken out and put back does. The previous round's tray is ignored until it leaves or changes.

6. Reading the evaluator window

On screen	Meaning
EVALUATION AREA box, green / grey	The station. Green = a tray is on it. Only batteries inside a tray on the station count.
STATION NOT FOUND (red)	The model never detected the station; the whole image is used. Check the camera view and lighting.
Green row and green marks	Evaluation is active for that round.
Black marks, OK / MISTAKE	Round evaluated; marks are frozen.
station: hand at the station	Battery cues stay on; the whole-tray reading for the verdict waits for the hand to leave.
GUI linked / GUI NOT FOUND	LSL connection to the GUI PC.
WARNING ... GUI shows ...	The log file on the camera PC does not match what the GUI displays (different repo version). The evaluator then uses what the GUI shows.

7. Settings you may want to tune

Setting	Where	Default	Effect
CUE_FAST_CONFIDENCE	live_evaluator.py	0.75	Model confidence from which a battery cues on the first frame. Higher = fewer false cues, slower.
--cue-frames / CUE_STABLE_FRAMES	live_evaluator.py	2	Frames a less certain battery must be seen before its cue.
--no-roi, --roi-scale	live_evaluator.py	area on, 1.0	Model only on the area around the station (yellow MODEL AREA). Scale below 1 = faster, above 1 = finer.
--exposure, --fps	live_evaluator.py	live_test.py values	Camera exposure (microseconds) and frame rate. Shorter exposure and more fps = faster cues, needs more light.
--stable-frames / STABLE_FRAMES	live_evaluator.py	8	Frames the whole tray must be unchanged (no hand) to count for the verdict.
--tray-gone / TRAY_GONE_FRAMES	live_evaluator.py	10	Frames the tray must be off the station before the round is judged.
STATION_MARGIN	live_evaluator.py	0.10	How far the tray centre may sit outside the station box.
--predict-only	live_evaluator.py	off	Never send verdicts; only predict and log.
correct_cue_audio, incorrect_cue_audio, CUE_AUDIO_MAX_SECONDS	config.py	see table	Cue sounds and their 1 s limit. Change the file name to change the sound.
HAPTIC_*_PATTERN / _POSITION	config.py	pos. 3 back, neg. 2 front	Vest cue patterns and sides.
USE_REMOTE_EVALUATOR	config.py	True	False = no camera link, manual y / c evaluation again.

8. Troubleshooting

Problem	Likely cause and fix
GUI NOT FOUND / no "Connected" line	Different networks, VPN on, or firewall blocking Python on one PC. Both programs must be running. Test with `--no-camera`.
Connected, but no cues at the participant	Condition is baseline / training / robot (no cues by design); bHaptics Player not running; headphones not found (GUI console prints the participant audio device at start); `websocket-client` missing.
Cues come for the wrong answers	Camera PC has a different `log_files/` version (look for the WARNING line), or slots are mirrored: slot 1 must be the left third of the tray in the camera image.

Problem	Likely cause and fix
Cues fire while a battery passes over another slot	Raise `CUE_FAST_CONFIDENCE` (0.85-0.9) and `--cue-frames` (3-4).
Cues feel late	Check the footer of the evaluator window (model fps, ms) and the CamMs / LinkMs numbers in the `Battery_Cue_...` markers. Shorten camera exposure / raise frame rate, lower `--roi-scale`, use wired headphones (Bluetooth adds roughly 0.15-0.25 s).
Round judged MISTAKE although the tray was right	A battery was not detected before the tray left: check model confidence and lighting; raise `--tray-gone` if the tray is judged too early.
No verdict arrives	The tray is still on the station (the round is only judged when it leaves; the handover is the safety net); image task rounds are not evaluated; or `--predict-only` is on.
Evaluator started late, shows s?? / no answer file	Restart it with `--subject N --condition NAME`, or go back to the GUI setup screen and start again.

9. Files and markers

What	Where
Round log of the GUI (score, verdict, times)	GUI PC: `logs/sXX_<condition>_<time>.csv`
Evaluator log (expected vs read, trigger, battery cues with times)	Camera PC: `logs/evaluator_sXX_<condition>_<time>.csv`
Questionnaires	GUI PC: `logs/questionnaires/`
Markers from the GUI	`ParadigmMarkers`: Dual_Workpiece_Task_Started_..., Battery_Cue_..., Voice_Deliver_..., Handover_Confirmed_..., Assembly_Evaluated_...
Markers from the camera PC	`BatteryEvaluator`: Eval_Battery_..., Eval_Prediction_..., Eval_Verdict_...
Real link delay	In the XDF: time between `Eval_Battery_...` and the matching `Battery_Cue_...`.

Testing the evaluator alone (no GUI): `python live_evaluator.py --test` lets you pick a log file; say "deliver" or press D to move to the next row.